

Problem

Design a problem to help other students better understand ideal transformers.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand ideal transformers.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A 4-kVA, 2300/230-V rms transformer has an equivalent impedance of $2 \angle 10^\circ \Omega$ on the primary side. If the transformer is connected to a load with 0.6 power factor leading, calculate the input impedance.

Solution

$$Z_{in} = Z_p + Z_L/n^2, \quad n = v_2/v_1 = 230/2300 = 0.1$$

$$v_2 = 230 \text{ V}, \quad s_2 = v_2 I_2^*$$

$$I_2^* = s_2/v_2 = 17.391 \angle -53.13^\circ \text{ or } I_2 = 17.391 \angle 53.13^\circ \text{ A}$$

$$Z_L = v_2/I_2 = 230 \angle 0^\circ / 17.391 \angle 53.13^\circ = 13.235 \angle -53.13^\circ$$

$$\begin{aligned} Z_{in} &= 2 \angle 10^\circ + 1323.5 \angle -53.13^\circ \\ &= 1.97 + j0.3473 + 794.1 - j1058.8 \end{aligned}$$

$$Z_{in} = \mathbf{1.324 \angle -53.05^\circ \text{ kohms}}$$